Contributions and evidence boundaries

Contributions and evidence boundaries I

We make ten contributions, each paired with a theorem, figure, table, or generated token and with an explicit boundary on what the evidence establishes. They fall into four groups: the first two build the core and its recovery contract; the next two report and statistically qualify the contaminated-consensus result; the fifth and sixth make the robustness axes explicit and add the objective-backed server rule; and the remaining four are scoped extensions — parameter recovery, the tempered aggregation family, an aggregation-API transfer, and a disjoint-observation communication test.

Contributions and evidence boundaries I

1. **A discrete-categorical FedGVI core.** A typed, deterministic, pure-NumPy/ SciPy reimplementation of the FedGVI (Mildner et al. 2025) generalized-Bayes primitives — divergences, bounded robust losses, the generalized posterior, the cavity/factor algebra, and robust aggregation — in the discrete-categorical setting that active inference (Da Costa et al. 2020) uses. The objective (eq. 1) and its closed-form tempered-softmax solution are stated in Definition 1 and tested by the recovery-limit probes. The whole core is zero-mock and reproducible (Peng 2011).

Contributions and evidence boundaries II

2. **A recovery-tested connection between categorical pooling and robust Bayes.** A recovery certificate showing that the client KL/negative-log-likelihood loss limits recover Bayes and the server's zero-robustness branch recovers the project log-linear pool. Under the explicit shared-support, posterior-log-potential, and fixed-weight assumptions in sec. 5.8, that pool specializes Friston et al.'s Eq. 7 message-combination term (Friston et al. 2024), not the complete source protocol. Three recovery limits are pinned to bit-level residuals: the bounded losses recover NLL/Bayes (Corollary 6 + Proposition 4; ≤ 0 and ≤ 0 maximum residual, eq. 6), the Rényi divergence recovers KL (Lemma 3; ≤ 0 residual, eq. 3), and the server-side reweighting pool recovers the naive pool in its trusting limit (Theorem 5; ≤ 0 residual, eq. 7). Robustness is thereby a tested recovery-limit extension, not a replacement.

Contributions and evidence boundaries III

3. **End-to-end evaluation of robust federated active inference.** Three worked categorical source-mechanism analogues — communicating colonies reaching lower free energy (sec. 7.2), Dirichlet language acquisition (sec. 7.3), and structure emergence by Bayesian model reduction (Friston and Penny 2011) (sec. 7.4) — plus a contaminated-sentinel robustness sweep (sec. 7.5) in which the naive pool degrades and at least one server-side robust member clears the configured threshold at the most severe swept rate.

4. **A statistically qualified server-side contrast.** The "robust beats naive" conclusion for the declared contamination rate is produced only by a matched-pairs Wilcoxon signed-rank test (Wilcoxon 1945) deflated across the divergence family with Benjamini–Hochberg FDR (Benjamini and Hochberg 1995), reported with bootstrap confidence intervals (Efron and Tibshirani 1993) and observed-effect design-power planning. Across 960 paired trials the headline display method (RKL; tied set: RKL, AR, beta, rcce) reaches accuracy 0.9867 against the naive pool's 0.9021 at the verdict rate ($q = 1.11 \times 10^{-158}$, rank-biserial-derived d-equivalent $=$ saturated (r=+1)). The predeclared selection rule is largest positive rank-biserial effect_size; stable method order tie-break; the method with the largest paired mean difference is AR. Every headline number is a generated token.

Contributions and evidence boundaries IV

5. **An explicit accounting of the robustness axes.** A clear separation (sec. 3.3) between the client-side per-agent update, which inherits FedGVI's bounded-influence result under its stated source assumptions; the sharp server-side reweighting heuristic, whose positive formal property is its recovery limit and whose declared separable objective class has a scoped no-go result; and the conservative variational server rule, which is objective-backed and has a redescending effective-weight update but is not the accuracy-maximizer. Downstream users are told exactly which result is theoretically backed and which is a labeled heuristic.

Contributions and evidence boundaries V

6. **An objective-backed server aggregator with redescending weights.** We derive an aggregation free energy eq. 8 whose exact block updates define the `variational_aggregate` rule (sec. 5.8.2, sec. 11): each exact block update monotonically decreases a stated objective, a converged fixed point is coordinatewise stationary, and the implementation keeps the lowest observed objective among converged configured starts (or reports the best unfinished trace as non-converged), recovers the standard log-linear pool in the trusting limit (eq. 7), and — unlike the sharp heuristic — carries a proven raw effective-weight bound (fig. 14, fig. 15). The honest cost, stated plainly, is conservatism: it is a maximum-entropy-biased consensus and trades peak point-accuracy for that control, so it complements rather than replaces the sharp heuristic of contribution 5.
7. **Executed finite-grid acuity-recovery experiment.** At each value in the 0.60, 0.70, 0.80, 0.90 acuity grid, the study generates 200 synthetic observations in each of 960 trials and selects acuity by marginal-likelihood grid search over the declared finite grid. The observed mean absolute error is 0.0232 with $R^2 = 0.9999$ (fig. 28). Acuity-by-colony-size behavior belongs to the separate sensitivity study; it is not a parameter-recovery result.

Contributions and evidence boundaries VI

8. **The $\mathbf{F}_\lambda$ tempered aggregation family.** A one-parameter $\lambda > 0$ generalization of the variational aggregate (sec. 11.5): $F_\lambda(q, a) = \sum_n a_n \cdot \mathrm{CE}(q, q_n) - \lambda H(q) + (1/c)\,\mathrm{KL}_{\mathrm{gen}}(a \| w)$. At $\lambda = 1.0$ the temperature is unity and the objective reduces to the standard variational aggregate bit-for-bit. Lower λ sharpens the variational q-block toward a maximizing state; it does not algebraically recover `robust_aggregate`. The raw effective-weight update and its bound are preserved for all λ. Full derivation in sec. 11.5.
9. **An aggregation-API transfer demonstration.** The same `robust_aggregate` API that governs the POMDP studies is exercised unchanged with one deterministic MLP trained with the density-power β-loss (16 hidden units, $\beta = 0.5$; generalized variational inference with a point-mass variational family) as the per-client model, supporting portability of the server API to this additional model class (sec. 7.6) when the optional `torch` extra is installed (sec. 5.26); without it the MLP run is skipped and its tokens render accordingly.

10. **Communication benefit under disjoint observations.** A multi-agent extension (sec. 13.7.1) in which 3 agents each observe a 2-slot disjoint window shows that belief sharing materially improves over isolated-agent accuracy in the declared configuration — isolated agents clear the 0.167 chance baseline but stay far below the communicating consensus, which itself remains well short of full accuracy: across 128 seeds isolated accuracy is 0.326 versus communicating 0.493, a reproducible margin under the declared matched-seed comparison (Wilcoxon $p = 0.0000$); this is evidence for the configured disjoint-observation protocol, not a universal communication theorem.

The remainder of the paper proceeds as follows. sec. 5 develops the FedGVI core and the recovery limits; sec. 6 states the numbered recovery theorems and the expected-free-energy identity; sec. 5.22 fixes the configuration; sec. 7 reports the 9 studies, beginning with the recovery checks (sec. 7.1); sec. 8 and sec. 9 synthesize; and sec. 10 and sec. 8.15 document determinism, scope, limitations, and the standing of each robustness axis.